

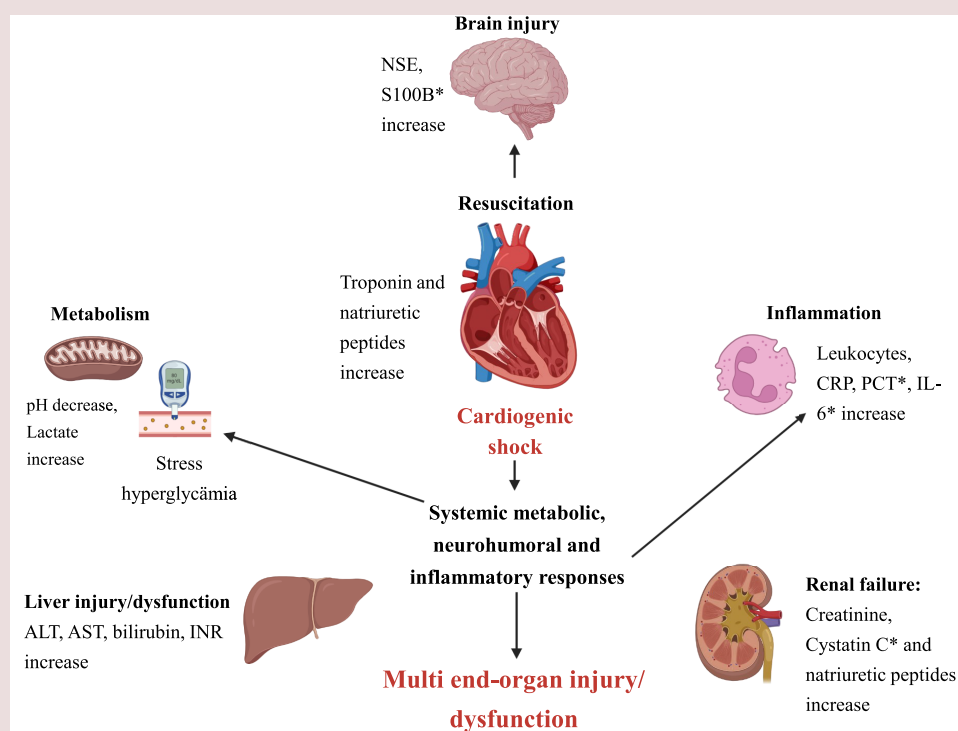
Best of cardiovascular biomarkers in 2025

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Received 11 November 2025; accepted 11 November 2025; online publish-ahead-of-print 1 January 2026

Graphical Abstract



Conventional laboratory parameters in cardiogenic shock. The main limitation of all conventional biomarkers is lack of specificity for cardiogenic shock. The severe state of hypoperfusion in cardiogenic shock leads to pronounced systemic metabolic, neurohumoral and inflammatory responses. Cardiac troponin and natriuretic peptide increases reflect myocardial injury and stress. Longer periods of hypoperfusion in cardiogenic shock result in multi end-organ damage. Liver and kidney injury and dysfunction laboratory parameters are used for monitoring. After cardiogenic shock-related resuscitation also brain injury biomarkers (e.g. NSE, S100B) are important for assessing prognosis. Temporal changes in biomarkers may guide prognostication.

*Alternative biomarkers without clearly proven benefit over standard markers but with potential advantages in specific clinical situations
Abbreviations: alanine amino-transferase (ALT), aspartate amino-transferase (AST), ‘neuron-specific’ enolase (NSE), C-reactive protein (CRP), interleukin (IL), international normalized ratio (INR), procalcitonin (PCT)

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Laboratory parameters are readily available, valuable tools for the diagnosis, prognostication, and management of acute cardiovascular diseases.¹ In 2025 original science published in the *European Heart Journal Acute Cardiovascular Care* again has significantly advanced our knowledge of cardiovascular biomarkers in this field. Research on blood-based biomarkers in cardiogenic shock was a key area of focus. As in the previous year,¹ cardiac troponin (cTn) testing at the point-of-care (POC) for the early diagnosis of acute myocardial infarction (MI) using high-sensitivity (hs) assays, diagnosis of acute heart failure, and risk stratification remained important topics.

Cardiogenic shock

Cardiogenic shock is caused by severe cardiac dysfunction, most frequently related to MI. It leads to a severe state of systemic hypoperfusion with neurohumoral and systemic inflammatory responses, which may result in multi-organ injury. Thus, markers of organ dysfunction and damage, metabolism and inflammation are potentially useful for diagnosis and risk stratification (see [Graphical abstract](#)). Lactate is an essential part of the classification and characterization of cardiogenic shock and used to monitor response to treatment. Cardiac biomarkers (troponin, natriuretic peptides) and markers of renal and hepatic failure are also already routinely used in practice.

Several published studies investigated the usefulness of these and other biomarkers in the context of clinical risk scores for predicting outcome in cardiogenic shock.^{2–6} Schupp et al.² investigated the predictive value of C-reactive protein (CRP)—an established biomarker of systemic inflammation—in cardiogenic shock following MI in the ECLS-SHOCK trial. Admission concentrations >61 mg/L were associated with higher all-cause mortality at 30 days compared with patients with values <5 mg/L. However, the inclusion of CRP concentration on admission to the IABP-SHOCK II score did not significantly improve prediction of outcome (area under receiver operating characteristic [AUROC] curve; 0.74 [95% confidence interval 0.68–0.79] vs. 0.71 [0.66–0.77]). This score already includes lactate, glucose (marker of stress hyperglycaemia) and creatinine concentrations as biomarkers. Interestingly, the maximum CRP concentration and white blood cell count on admission were not prognostic.² Peters et al.³ reported the development and external validation of a risk score in MI-related cardiogenic shock for the prediction of 30-day mortality. The model used variables available on admission including lactate, glucose and estimated glomerular filtration rate (eGFR) as significantly predictive biomarkers. In addition, the model based on variables available immediately after primary percutaneous coronary intervention included haemoglobin. The prognostic performances of both models were comparable (AUROC 0.72 vs. 0.73) in the external validation cohort, and comparable to the AUROC of the IABP-SHOCK II score with and without including CRP.² In a retrospective, observational study, Bjorn et al. confirmed the prognostic significance of the development of acute kidney injury based on RIFLE criteria (prevalence 44%; 25% of these patients required renal replacement therapy) for short- and long-term outcomes in MI-related cardiogenic shock.^{4,5} Similarly, liver injury, quantified as MELD-albumin score, which includes blood concentrations of bilirubin, creatinine and albumin, was significantly associated with short- and long-term all-cause mortality in MI-related cardiogenic shock.⁶ However, the predictive performance of this score for 30-day mortality, which excludes the International Normalized Ratio (INR) to avoid the effects of anticoagulants, was only modest (AUROC 0.63). This score has primarily been developed to assess chronic liver injury, and it is unlikely to be useful in clinical practice in cardiogenic shock patients. However, these results highlight the potential value of incorporating liver injury markers for risk stratification in cardiogenic shock. As in acute kidney injury, the likely

underlying pathophysiology includes a combination of arterial hypoperfusion and venous congestion.

In patients with MI-related cardiogenic shock who require resuscitation, biomarkers of anoxic brain injury may inform prognosis. Alaoui-Ismaili et al. confirmed the limited specificity of ‘neuron-specific’ enolase (NSE) for brain injury.⁷ This enzyme may also be released from damaged erythrocytes.⁷ Therefore, haemolysis must be considered as a cause of increased NSE concentrations, particularly in cardiogenic shock patients receiving circulatory support with a microaxial pump or extracorporeal membrane oxygenation. In such patients, the glial biomarker S100B, a calcium-binding protein found in astrocytes, may be a more accurate alternative biomarker of anoxic brain injury (see [Graphical abstract](#)).

Investigating differences in proteomic profiles may provide new insights into the complex pathophysiology of cardiogenic shock. In a pilot study Patel et al.⁸ used targeted proteomic profiling to identify promising novel biomarkers for the diagnosis of cardiogenic shock. Four biomarkers, soluble ST2 protein (sST2), fibroblast growth factor-23 (FGF)-23, cathepsin (CTS)-D, and growth differentiating factor (GDF)-15, were identified as potentially useful biomarkers to distinguish patients with cardiogenic shock from those with isolated heart failure or hypotension unrelated to cardiac diseases. sST2 had the greatest association with cardiogenic shock. These candidates require further evaluation, but novel techniques may eventually lead to the identification of new, clinically useful biomarkers for the diagnosis of early-stage cardiogenic shock or for risk stratification once the diagnosis is established.

Diagnosis of acute myocardial infarction and acute heart failure

Articles on the diagnosis of MI and acute heart failure confirmed and extended study results published previously.¹ In a multicentre observational trial, Greenslade et al.⁹ reported zero-hour thresholds to risk stratify patients with possible MI using a novel hs-cTnI POC assay on presentation to the Emergency Department (ED). The performance of the POC assay was similar to central laboratory hs-cTnI assays,⁹ and this assay also demonstrated a comparable performance with central laboratory assays when incorporated into a 0/2 h accelerated diagnostic pathway.¹⁰ Whilst encouraging, implementation studies that optimize the care pathway are required to ensure that introducing POC hs-cTn testing has a meaningful impact on length of ED stay.¹¹

Obergassel et al. investigated the sensitivity of multiple hs-cTn assays for the detection of early acute myocardial injury in patients undergoing pulmonary vein isolation as a clinical model of acute myocardial injury.¹² Following the procedure 78% of patients had evidence of myocardial injury with increases in cTnI that were 2 to 6-fold higher than for cTnT, depending on the hs-cTnI assay evaluated. These observations suggest differences in the epitopes targeted by cTnI assays may influence sensitivity, and that the ratio of cTnI to cTnT may be useful to differentiate acute from chronic myocardial injury. However, the significance of these findings are difficult to assess as cTn assays are neither standardized nor harmonized, and it is not clear whether the findings are due to differences in analytical characteristics of cTn assays or differences in the release kinetics of cTnI and cTnT from acutely injured myocardium.

Patients presenting with acute dyspnoea remain another frequent challenge in the ED. Doudeis et al.¹³ reported the development and external validation of a machine learning based decision-support tool that combines natriuretic peptides (either B-type natriuretic peptide or midregional pro-atrial natriuretic peptide) with clinical variables (i.e. age, sex, renal function, atrial fibrillation and other comorbidities) to report the probability of acute heart failure in an individual patient.

This tool was more accurate compared with using guideline-recommended natriuretic peptide decision thresholds for classification as has already been shown for N-terminal proBNP.¹⁴ This study demonstrates that machine learning can improve the diagnostic accuracy for acute heart failure of all natriuretic peptides and that the use of these tools in practice could provide more consistent performance across patient groups stratified by age, sex and comorbid conditions known to influence natriuretic peptide concentrations.

Summary

In cardiogenic shock the search continues for additional biomarkers that improve diagnosis and risk prediction over and above routinely measured markers, such as lactate, glucose, and creatinine. Currently available routine biomarkers lack specificity and cannot reliably differentiate early-stage cardiogenic shock from acute heart failure or systemic hypotension unrelated to an acute cardiac disease. New tools that incorporate markers of inflammation and brain injury provide only a modest improvement in the prediction of 30-day mortality. Novel approaches using proteomic techniques show promise to identify relevant pathophysiological pathways where new biomarkers could advance diagnosis and risk prediction for patients with cardiogenic shock. In parallel, we summarize new insights from biomarker studies that could enhance the utility of existing markers of myocardial injury and haemodynamic stress to accelerate diagnosis, differentiate acute from chronic disease, and improve the early recognition of both acute MI and decompensated heart failure.

Author contributions

Johannes Mair (Conceptualization, Visualization, Writing—original draft) and Nicholas Mills (Writing—review & editing)

Funding

NLM is supported by a Research Excellence Award (RE/24/130012), Programme Grant (RG/F/25/110169) and a Chair Award (CH/F/21/90010) from the British Heart Foundation and reports honoraria from Abbott Diagnostics, Siemens Healthineers, and Roche Diagnostics in the last 36 months.

Conflict of interest: JM reports research collaboration on cardiac biomarker point-of-care diagnostics with Siemens Healthineers, The Netherlands, in the last 36 months.

Data availability

There is no new data associated with this article.

Appendix

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